Intermediate — a file-upload app — Hints

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Hint 1 — one widget, bytes out

Every framework's file input hands you bytes:

- **Streamlit:** `uploaded = st.file_uploader("File", type=["txt","pdf"])` returns a file-like object (or `None`); `uploaded.read()` → bytes.
- **FastAPI:** `async def upload(file: UploadFile): data = await file.read()` — plus `file.filename`.
- **Flask:** `request.files["file"].read()`.

For a `.txt`, `data.decode()` gives the text; feed it to your pipeline.

Hint 2 — PDFs, and validating the upload

A PDF isn't text — save the bytes to a temp file and use Doc08's loader:

```
import tempfile
from langchain_community.document_loaders import PyPDFLoader
with tempfile.NamedTemporaryFile(suffix=".pdf", delete=False) as tmp:
    tmp.write(uploaded.read()); path = tmp.name
pages = PyPDFLoader(path).load()   # then join page.page_content
```

Always validate first: check the filename extension (`name.lower().endswith((".txt",".pdf"))`) and reject anything else with a clear error — never process an unknown or empty upload. Test the FastAPI version with `TestClient` and `files={"file": ("note.txt", b"...", "text/plain")}`. The full code is in the Solution.